LISSI Laboratory (EA 3956) — Université Paris-Est Créteil

End-to-End Integration of a VLM with a Neuro-Symbolic State Manager for Embodied Robot Manipulation on TIAGo

System Integration Demonstration

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Embodied Agent ▪ VLA ▪ Neuro-Symbolic ▪ Open-Vocabulary Perception ▪ TIAGo

Outline

1. Motivation
2. System Architecture
3. VLM Reasoner
4. Semantic World Model
5. Grasping Pipeline
6. Handover & Face Recognition
7. Discussion
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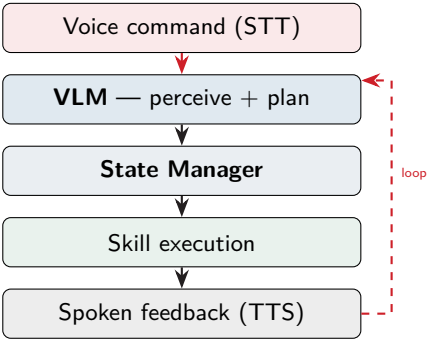
Building an Embodied Robot Agent

Goal: a robot that listens, sees, decides, and acts — in a **closed loop**, without retraining.

Three gaps in existing systems:

1. **Fixed-vocabulary perception**
   YOLO needs retraining for every new object class
2. **Static world models**
   RDF/OWL graphs built offline; cannot update after manipulation
3. **No GPU-free full loop**
   No open system closes voice→grasp→handover on a real mobile robot without a GPU

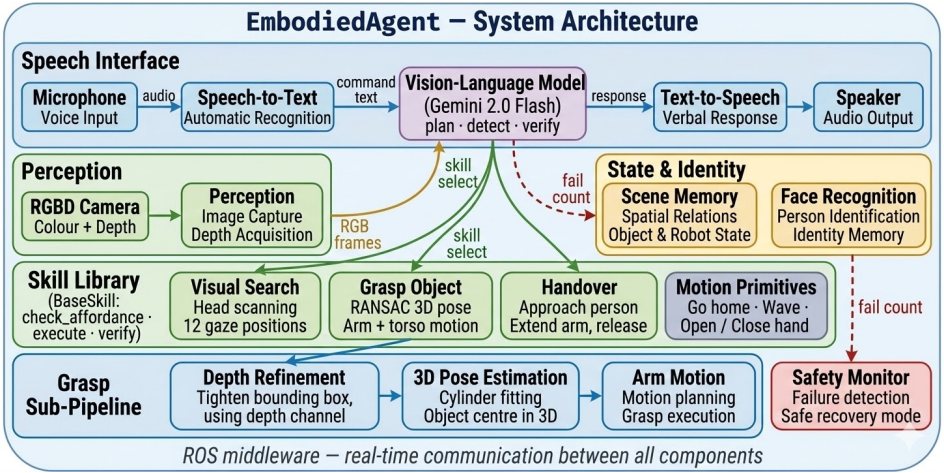


Our answer

Cloud VLM as cognitive core + live symbolic state manager, deployed on a real **TIAGo** at LISSI Lab, UPEC.

Three-Layer Neuro-Symbolic Architecture

L1 Neural (VLM: perceive + plan) $\longrightarrow$ **L2 Symbolic** (StateManager: scene ontology) $\longrightarrow$ **L3 Action** (Skill Library + ROS/MoveIt) $\longrightarrow$ **TIAGo hardware**



Seven Skills in the Action Library

Complete skill repertoire

Skill	Purpose
`search_with_head`	Scan 12 positions for target
`grab_bottle`	RGBD-guided grasp
`handover_to_person`	Arm extension + delivery
`wave`	Social gesture
`go_home`	Safe arm retraction
`open_hand`	Open gripper
`close_hand`	Close gripper

Continuous agent cycle

1. Transcribe voice (Whisper + cloud fallback)
2. VLM perceives scene & selects skill
3. StateManager updates world model
4. Skill executes on TIAGo (ROS/MoveIt)
5. VLM verifies before/after images
6. Spoken feedback via PAL TTS

↻ Repeat

Safety: 3 consecutive failures trigger safe-mode halt.

VLM Reasoner — One Call Does Everything

Key idea: a single Gemini 2.0 Flash call handles both tasks at once:

1. **Open-vocabulary detection**
   Detect any object by name — no fixed class list, no retraining
2. **Skill selection**
   Choose the action to execute and explain why

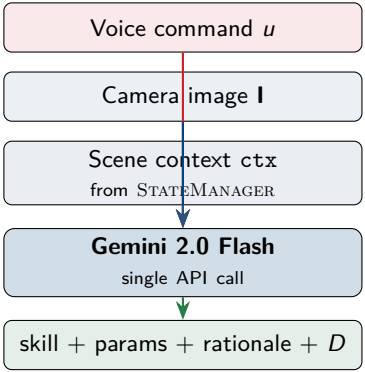
Single call returns:

$\mathcal{V}(u, \mathbf{I}, \text{ctx}) \rightarrow \{\texttt{skill}, \texttt{params}, \texttt{rationale}, D\}$

D = detected objects with bounding boxes

`rationale` = chain-of-thought reused at grasp time

No separate detector + planner pipeline.
No local GPU. Standard laptop CPU only.



StateManager — Live Semantic Belief State

Problem: a cloud VLM is completely *stateless* — it forgets everything between calls.

Solution: StateManager persists across the session:

- Tracks every detected object (bbox, position, count)
- Computes spatial relations from bounding boxes automatically
- Records gripper state and task history
- **Serialises everything as text prepended to each VLM prompt**

Effect

A stateless cloud API becomes a **stateful, session-aware agent.**

Context injected into every VLM prompt:

```
Gripper: holding:bottle
Location: table_area
Relations: bottle left_of cup,
   cup near box
Memory: cup (3x, 12s ago)
History: search_with_head OK
```

Auto spatial relation rules

dist < 150 px	near
centroid$_1$ left	left_of
bottom edges ± 50 px	on_table
age > 5 min	excluded from context

Active Visual Search

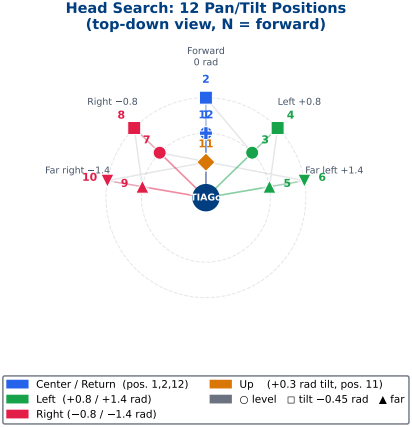
Problem: the target object may not be visible in the robot's initial field of view.

Solution: `search_with_head` — systematically scan before attempting any grasp.

Scan pattern

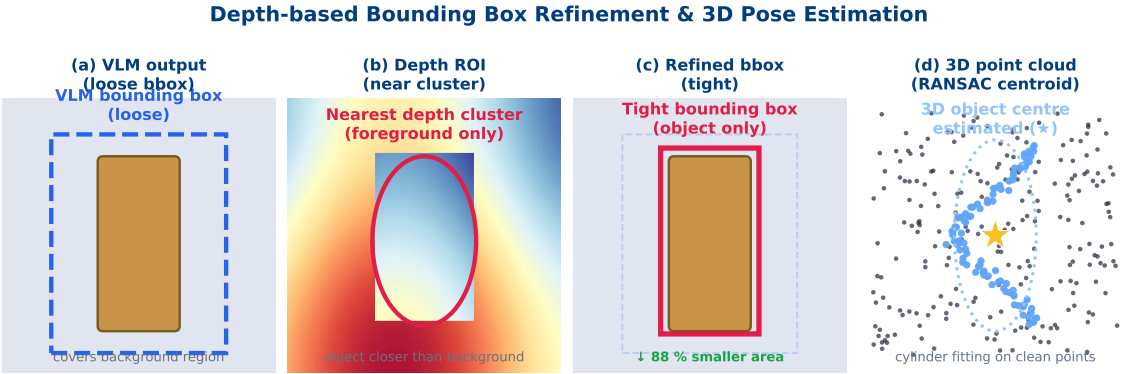
- **12 pan/tilt positions** cover the workspace
- Pan: -1.4 to $+1.4$ rad (left/right)
- Tilt: -0.45 to $+0.3$ rad (table/shelves)
- VLM queries for target at each stop
- **Search ends as soon as target is found**

For person targets: face-visibility check prevents false positives on feet or torsos.



Depth-Based Bounding Box Refinement

Problem: VLM boxes are semantically correct but spatially coarse (include table & background). Raw box ⇒ wrong 3D pose ⇒ arm misses. **Fix:** depth-filter to nearest foreground cluster ± 12 cm.



3D Pose Estimation and Arm Execution

Pose estimation pipeline

1. **Back-projection:** depth + intrinsics (f_x, f_y, c_x, c_y) + TF to base frame
2. **RANSAC cylinder fit:** centre (x, y) and radius — works for bottles and cans
3. **Centroid fallback:** median of front-face cluster for flat or box-shaped objects
4. **Stability:** 5 estimates averaged; reachability check removes out-of-reach poses

Arm execution — top-down grasp

1. MoveIt plans collision-free path **20 cm above** object
2. Torso lift joint descends **straight down** (no IK — guaranteed vertical approach)
3. Gripper closes; trunk rises; arm goes home

Context-grounded VLM detection
↓
Depth bbox refinement
↓
Back-projection to 3D
↓
RANSAC cylinder / centroid
↓
5-frame average pose
↓
MoveIt + torso descent

No GPU. No segmentation model. Standard laptop CPU.

Handover and Face Recognition

Handover skill

1. Return home; clear stale MoveIt planning scene
2. VLM detects person in current RGB frame
3. Depth back-projects person centroid to 3D
4. MoveIt extends arm at **0.85 m height**, up to **0.62 m** toward person
5. TTS: "Please take the object from my hand"
6. Wait **4 s** → open gripper → go home

Person collision box added during planning — arm cannot hit the person.

FaceManager — identity recognition

- Separate HTTP microservice (decoupled from ROS)
- Encodes faces as compact embedding vectors
- Matches by Euclidean distance below fixed threshold

Social interaction

- **Known person** → robot greets by name
- **Unknown face** → robot asks name via STT, VLM parses response, face stored for session

Key Insights

Neuro-symbolic agent loop

VLM: rich, open-vocabulary observations.
SM: structured, interpretable world model.

Together: stateless cloud API becomes a **stateful, session-aware agent.**

Live vs. static ontology

Unlike offline RDF graphs, SM updates at every inference step from live detections.

Once an object is grasped, the state change propagates immediately — re-grasping is impossible.

Unified VLM = lightweight planning

Single model for detection + skill selection eliminates the detect-then-plan error cascade.

`rationale` acts as chain-of-thought seeding the grasp-level prompt — hierarchical planning without a separate planner.

Computational footprint

No local GPU. RANSAC, MoveIt, ROS all on a standard laptop CPU.

Trade-off: Gemini API outage = system failure.

Limitations and Future Work

Current limitations:

- **Geometry:** RANSAC cylinder fails on flat, irregular, or handle-bearing objects
- **Cluttered scenes:** nearest-depth cluster can merge adjacent objects at the same depth
- **Cloud dependency:** network outage stops the system
- **Static locations:** navigation targets hardcoded; no semantic 3D map yet
- **Evaluation:** controlled lab only — unstructured environments not yet tested

Future directions:

- Superquadric fitting or learned grasp templates for non-cylindrical objects
- Instance segmentation for cluttered scenes
- Local VLM (LLaVA-7B) for offline resilience
- SayPlan-style semantic 3D map for multi-room deployment
- Task-adaptive memory timeout (vs. fixed 5 min)
- Structured experiments: success rates, ablation of depth refinement vs. raw VLM box

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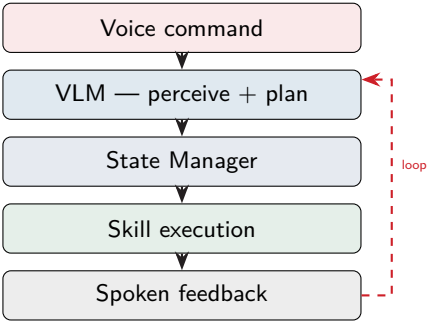
Conclusion

What we demonstrated on real TIAGo:

1. **Closed perception-reasoning-action loop**
   Voice command → physical handover, without a GPU
2. **Live semantic state injection**
   Spatial relations inferred every step; multi-instance disambiguation without weight updates
3. **Unified VLM** (Gemini 2.0 Flash)
   Open-vocabulary detection + skill selection in a single API call
4. **Depth foreground extraction**
   Tight 3D localisation from loose VLM boxes, no segmentation model

A practical blueprint for language-guided manipulation on commodity service robots.



No GPU No retraining Real TIAGo

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Thank You

Questions?

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